# POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2016
Programme: BE | Full Marks: 100
Course: Microprocessors | Pass Marks: 45
Time : 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.*
*The figures in the margin indicate full marks.*
*Attempt all the questions.*

1. a) Define Microprocessor. Compare Intel 8085, 8086 and 80386 Microprocessors on the basis of their features and internal architectures. 7
   b) Explain flags in 8085 with the help of a diagram. What is the function of LDAXB? 8
2. a) Write an ALP in 8085 to find whether the given number is palindrome or not. 8
   b) Write a 8085 program to find smallest of three numbers. 7
3. a) Define T-state. Draw the labeled timing diagram for the instruction LXI B, C00FH. 7
   b) What are the three basic functions microprocessor should satisfy to interface any memory chip? Explain how 2048×8 EPROM is interfaced with 8085 microprocessor. 8
4. a) What are interrupts? Differentiate between I/O mapped I/O and Memory mapped I/O. 7
   b) Draw and explain the functional block diagram of 8259A PIC. 8
5. a) Write an assembly language program for 8086 to find the sum of two numbers input by the user through the keyboard and display the sum on the screen. 8
   b) Write an 8086 ALP for MASM in DOS-BIOS mode to display the string "Pokhara University" on screen without using 09H. 7
6. a) Explain procedures and macros. Why IVT is used in microprocessor. 7
   b) Draw and explain the connections to show how RS-232C is made compatible with TTL logic along with its connections for null modem. 8
7. Write short notes on: (Any two) 2×5
   a) HOLD and HLDA pin
   b) BISYNC Protocol
   c) DMA



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